

Pair-Order Consistent and Uncertainty-Aware Semantic Change Detection from Paired RGB Frames

Ekanayake T.N.D.S.W., Lelwala J.U.P., Pabasara W.G.K.,
Bulagala D.W.K.G., Rajapaksha T.N.D.W.

Department of Computer Science and Engineering, University of Moratuwa
Moratuwa, Sri Lanka

Abstract

Manual comparison of video versions is slow, while pixel subtraction produces false alarms under camera displacement, lighting, colour grading, blur, and compression. This proposal develops a resource-conscious Siamese change detector that localizes meaningful edits between corresponding RGB frames from reference and comparison versions. FC-Siam-Diff is the mandatory CNN baseline; a Siamese ResNet-18 is a stronger comparison. The proposed extension combines signed pairwise fusion, bounded feature alignment, and confidence-aware prediction. It will be trained and benchmarked on public paired-image datasets and tested on an authorised target-domain video set using scene-disjoint splits, nuisance-only hard negatives, localization metrics, false-alarm measures, calibration, and hardware efficiency. The scope is deliberately limited to a reproducible binary change detector, with directional labels treated as an extension when annotations are available.

Keywords

semantic change detection, Siamese network, FC-Siam-Diff, ResNet, feature alignment, pair-order consistency, uncertainty

1 Problem Motivation

The real-world problem is to compare a frame I_1 from a reference video version with its corresponding frame I_2 from a comparison or modified version at the same semantic time, then identify meaningful editing changes at pixel level. The inputs are not normally adjacent frames from one ordinary sequence, because ordinary inter-frame actor and camera motion would dominate the desired edit signal. In video post-production, editors and quality-control teams must find missing, added, moved, or altered objects across long sequences. Manual inspection is repetitive and costly. A raw RGB or structural-difference map is fast, but it confuses semantic edits with camera motion, illumination changes, colour grading, shadows, blur, occlusion, and codec artifacts [12].

The project therefore treats comparison as dense change segmentation:

$$M = f(I_1, I_2), \quad M(x, y) \in \{0, 1\}, \quad (1)$$

where M highlights meaningful change and suppresses nuisance variation. The intended user is a human reviewer: the system should reduce search time and rank candidate regions, not make an unsupervised production decision. The main beneficiaries are editors, visual-effects and colour teams, archivists comparing versions, and researchers studying robust paired-image representations. The project is feasible on free GPU resources because it prioritizes lightweight CNNs over large foundation models.

Table 1: Provisional datasets and roles. Counts are subject to licence and quality checks.

Dataset	Data overview	Role and limitation
SYSU-CD [10]	20,000 aerial RGB pairs, binary masks	Baseline pretraining; domain gap from video.
LEVIR-CD [3]	637 high-resolution pairs, building masks	Boundary evaluation; near-registered aerial imagery.
PCD [8]	200 panoramic street-image pairs	Viewpoint/illumination robustness; small dataset.
Project video set	Matched frames from two authorised video versions	Optional fine-tuning; mandatory final test for post-production claims.

1.1 Research-Gap Synthesis

The reviewed approaches expose three gaps: fine-grained localization outside remote sensing, accurate high-resolution masks under limited compute, and learning with limited dense annotations. YOLO-style detectors are fast but do not naturally recover irregular pixel masks; Siamese CNNs are efficient; and transformers such as BIT and ChangeFormer add global context but generally require more memory and paired mask data [1, 2]. Given the semester scope, this project addresses the first two gaps through a lightweight Siamese detector, nuisance-aware training, and bounded feature alignment. A broad benchmark and weakly supervised learning remain future work.

2 Identified Datasets

The staged dataset plan and data characteristics are summarized in Table 1. Public datasets will support baseline training and architecture comparison, but their aerial or street-view domains cannot establish post-production usefulness. That claim will require a held-out authorised video set containing corresponding frames from reference and comparison versions, binary masks, nuisance labels, ignore regions, source-video identifiers, and timestamps or frame indices. Controlled repeated-take pairs may be included only when correspondence can be established. Splits will be made by source production, video, and scene before frame extraction, cropping, or augmentation so that adjacent or near-duplicate frames cannot cross partitions.

ChangeSim or CLEVR-Change may be used only for a controlled directional-label or nuisance-robustness experiment if access and annotation formats are confirmed. They will not replace the primary binary benchmark or the target-domain video test, and synthetic

results will be reported separately from real-video results. If the authorised video test cannot be completed, the project will be reported as paired-image change detection and will make no claim about post-production performance.

3 Baseline Model and Implementation

The mandatory baseline is FC-Siam-Diff [5]. It applies a shared encoder to both RGB images, computes multi-scale absolute feature differences

$$D^s = |E_\theta^s(I_1) - E_\theta^s(I_2)|, \quad (2)$$

and reconstructs a binary mask with a U-Net-like decoder [7]. A Siamese ResNet-18 model with the same decoder and training protocol is the stronger learned baseline. RGB/SSIM thresholding is a non-DNN sanity baseline. If compute permits, BIT will be an advanced transformer comparison [2].

Both frames will be resized to a fixed resolution and normalized identically. Geometric augmentations will be shared by both inputs; independent photometric perturbations will simulate illumination and colour variation. The loss will combine weighted binary cross-entropy and Dice loss. The initial tuning grid is learning rate $\{10^{-3}, 3 \times 10^{-4}, 10^{-4}\}$, frozen versus fine-tuned encoder, input resolution, and validation threshold. Checkpoints and thresholds will be selected using validation scenes only; test scenes remain sealed. Results will report precision, recall, F1, IoU, changed-pixel ratio, parameter count, training time, and end-to-end latency.

3.1 Baseline Evidence

Published FC-Siam-Diff results on WHU-CD and LEVIR-CD provide a reproducible reference point [9]; the initial ResNet-18 settings follow a related Siamese study [4].

Table 2: Reported FC-Siam-Diff baseline results [9].

Dataset/protocol	Precision	Recall	F1	IoU
WHU-CD [9]	0.856	0.747	0.798	0.663
LEVIR-CD [9]	0.846	0.892	0.871	0.772

Training will start from ImageNet-pretrained encoders with Adam and a learning-rate comparison of 10^{-3} , 3×10^{-4} , and 10^{-4} . The selected input resolution, encoder freezing schedule, and decision threshold will be fixed from validation scenes before the sealed test evaluation. Checkpoints, seeds, licences, and environment versions will be recorded for reproducibility.

4 Proposed Contribution

An ordinary Siamese encoder, feature differencing, or U-Net mask is established and is not claimed as novelty. The proposed contribution is a reliability-focused extension: a shared EfficientNet-B0/B2 or ResNet-18 encoder, signed and absolute feature fusion, and a bounded change-aware alignment module. The module estimates small feature offsets but uses a provisional change-confidence gate so that genuine moving or newly introduced objects are not aligned away. EfficientNet provides a controlled accuracy-resource comparison [11].

The project therefore targets the first two gaps: video-oriented nuisance robustness and accurate mask prediction under a modest latency/memory budget. Weak supervision and creation of a broad benchmark remain future extensions. Standard YOLO is a single-image detector rather than a paired-image change detector [6]. If compatible box annotations become available, an optional detect-and-match or six-channel early-fusion baseline will be evaluated using mAP and change-count error separately from mask-based F1/IoU.

When directional annotations are available, the decoder will additionally predict appeared and disappeared regions. A pair-order consistency loss will enforce that reversing (I_1, I_2) swaps these two channels while preserving the union change mask. This is a two-frame symmetry constraint, not a claim of multi-frame temporal modelling. Confidence calibration or selective prediction will allow the model to abstain on ambiguous nuisance-shifted pairs. The hypotheses are that alignment reduces nuisance-induced false positives, confidence gating protects real changes, and pair-order consistency reduces contradictory predictions. Improvement will be measured relative to FC-Siam-Diff and ResNet-18, not by an unsupported absolute target.

5 Planned Analysis, Outputs, and Resources

The planned data analysis compares RGB/SSIM, FC-Siam-Diff, and Siamese ResNet-18 under one common split before adding the proposed module. Ablations test fusion, alignment, confidence gating, and pair-order consistency. Results will be stratified by change size and nuisance severity (viewpoint, brightness, blur, occlusion, and compression). At least three seeds will be used for principal models where time permits.

Primary metrics are changed-class precision, recall, F1, IoU, and boundary quality. Robustness metrics are false-positive pixels and connected regions per no-change pair, image-level false-alert rate, and performance by nuisance condition. Reliability metrics are expected calibration error or Brier score and risk-coverage curves. Efficiency metrics are parameter count, pair-input FLOPs, peak GPU memory, and measured warm-up-excluded latency. Overall accuracy is secondary because unchanged pixels dominate.

The minimum viable implementation is RGB/SSIM, FC-Siam-Diff, Siamese ResNet-18, the lightweight alignment model, and a held-out video test for any post-production claim. Experiments will use Google Colab/Kaggle GPUs or personal hardware, PyTorch/TorchRS, version-controlled code, licensed storage, and a mask-annotation tool. Larger EfficientNet variants and BIT are stretch goals. Only authorised/licensed footage will be used; personal content will be minimized or blurred and publication examples will be cleared. The principal risks are target-domain annotation delay, domain mismatch, class imbalance, near-duplicate leakage, and alignment erasing real changes; source-level splits, hard negatives, weighted loss, a pilot annotation audit, and bounded alignment mitigate these risks.

Expected outputs include reproducible binary masks, benchmark and ablation tables, calibrated confidence scores, qualitative overlays, and nuisance-shift failure cases. The report will document when the detector should defer to human review.

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